

Huawei Solar Inverter Interface Specification

Professional Register Mapping & Industrial Control Plane Architecture Definitions (Modbus-RTU / Modbus-TCP Protocols) — Technical Reference Guide | Issue 04

1. Core Telemetry & Electrical Metrics (Real-Time Performance)

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Active power	32080	I32	2	1000	kW	RO	Real-time active output power tracking loop.
Reactive power	32082	I32	2	1000	kVar	RO	Real-time reactive power loops.
Input power	32064	I32	2	1000	kW	RO	Real-time total DC input power check metric.
Peak active power of current day	32078	I32	2	1000	kW	RO	Highest daily active generation capacity peak achieved.
Power factor	32084	I16	1	1000	N/A	RO	Inverter system power factor scaling quotient.
Efficiency	32086	U16	1	100	%	RO	Real-time DC-to-AC conversion efficacy calculation output.

2. Photovoltaic String Performance (DC Inputs)

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
PV1 voltage	32016	I16	1	10	V	RO	DC voltage metrics input loop for Photovoltaic String 1.
PV1 current	32017	I16	1	100	A	RO	DC current metrics input loop for Photovoltaic String 1.
PV2 voltage	32018	I16	1	10	V	RO	DC voltage metrics input loop for Photovoltaic String 2.
PV2 current	32019	I16	1	100	A	RO	DC current metrics input loop for Photovoltaic String 2.
PV3 voltage	32020	I16	1	10	V	RO	DC voltage metrics input loop for Photovoltaic String 3.
PV3 current	32021	I16	1	100	A	RO	DC current metrics input loop for Photovoltaic String 3.
PV4 voltage	32022	I16	1	10	V	RO	DC voltage metrics input loop for Photovoltaic String 4.
PV4 current	32023	I16	1	100	A	RO	DC current metrics input loop for Photovoltaic String 4.

3. Safety Diagnostics & Thermal Management

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Device status	32089	U16	1	1	N/A	RO	System state profile mapping index. 0x0000: Standby: initializing 0x0001: Standby: insulation check 0x0200: On-grid active run 0x0300: Shutdown: Fault.
Fault code	32090	U16	1	1	N/A	RO	Central diagnostic identifier mapping to unique device fault registers.
Internal temperature	32087	I16	1	10	°C	RO	Inverter system cabinet thermal ambient core monitoring loop.
Insulation resistance	32088	U16	1	1000	MΩ	RO	DC array ground fault barrier check metric. Safety threshold indicator.

4. Energy Accumulation & Yield Statistics

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Daily energy yield	32114	U32	2	100	kWh	RO	Yield generation accumulated within the current daily tracking loop.
Accumulated energy yield	32106	U32	2	100	kWh	RO	Lifetime absolute continuous active yield tracking summation standard registry.

5. System Temporal Context & Synchronization

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
System time	40000	U32	2	1	N/A	RW	Central clock context synchronization register.
Startup time	32091	U32	2	1	N/A	RO	Daily execution dawn initialization timestamp context (Epoch seconds).
Shutdown time	32093	U32	2	1	N/A	RO	Closing operational threshold suspension execution timestamp context (Epoch seconds).

6. Remote Control Commands & Execution Configuration

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Startup	40200	U16	1	1	N/A	WO	Write 1 to force manual absolute programmatic hardware runtime startup sequence trigger.
Shutdown	40201	U16	1	1	N/A	WO	Write 1 to force manual absolute programmatic hardware operational shutdown trigger.

7. Power Scheduling & Grid Compliance Parameters

Signal Name	Addresses	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Failsafe Active Power Limit [kw][High Accuracy]	42405	I32	2	1000	kW	RW	Power safety limit boundary baseline enforcement. Range context: [0.000, Pmax].
Fast power scheduling	45086	U16	1	1	N/A	RW	Enables or disables rapid grid telemetry scaling protocol (0: Disable 1: Enable).
Grid code	42000	U16	1	1	N/A	RW	National compliance safety boundary selector. Maps to Table 5-3 region maps (e.g., 0: VDE-AR-N-4105 Germany, 1: NB/T 32004 China, 2: UTE France, 13: CEI0-21 Italy, 19: AS4777 Australia).